

- i) Suppose that the data for analysis includes the attribute age. The age values for the data tuples 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

a) What is Mean and Median.

$$\text{Mean}(\bar{x}) = \sum_{i=1}^n \frac{w_i x_i}{w_i} \quad \left[\frac{1}{N} \sum_{i=1}^n \frac{w_i x_i}{w_i} \right]$$

$$= \frac{809}{27} = 29.96 = 29.96 = 30$$

$$\text{Median} (N + \frac{1}{2}) \quad 27 + \frac{1}{2} = 14$$

b) Mode and Modality.

25 and 35 are repeating for 4 times it is bi modal

c) Mid range.

$$\text{Mid rang:- } \frac{\text{Max} + \text{Min}}{2} = \frac{13 + 70}{2}$$

$$= 41.5$$

d) first Quantile and third Quantile.

Q_1 and Q_3

Q_1 = better middle values

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$$\text{Midvalue} = 20 = 7$$

$$Q_1 = 20$$

$Q_3 = 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70$

$$\text{Midvalue} = 35$$

$$Q_3 = 35$$

$$IQR = 15$$

$$Q_3 - Q_1$$

e) five-number summary

min, Q_1 , median, Q_3 , max

$$\text{Min} = 13$$

$$Q_1 = 20$$

$$\text{median} = 25$$

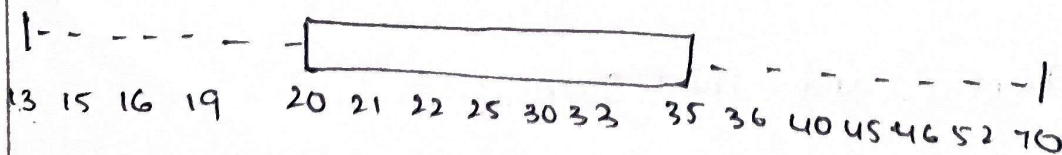
$$Q_3 = 35$$

$$\text{max} = 70$$

f) boxplot of the data

$$Q_1 = 20, Q_3 = 35$$

13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36, 40, 45, 46, 52, 70



2) Given the following Measurements for the variable - age:

$$\frac{1}{N} \sum_{i=1}^n \frac{w_i x_i}{w_i}$$

18, 22, 25, 42, 28, 43, 33, 35, 56, 28,

Standardize the variable by the following compute the mean absolute deviation of age.

$$\text{Mean} = \frac{330}{10} = 33$$

Age $(x - \bar{x})$

18 - 5

22 - 11

25 - 8

42 - 9

43 - 10

33 - 0

35 - 2

56 - 23

28 - 5

$$\text{MAD} = \frac{88}{10} = 8.8$$

3) Suppose that data for analysis includes the attribute age.

The age values for the data tuples are (increasing order)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36,

40, 45, 46, 52, 70 depth of 3.

Total values = 24

no bin = 3

depth = 9

Total no. of bins = $\frac{24}{3} = 8$ bins required

Bin	original values	Bin Mean	smoothed value
B ₁	13, 15, 16	14.67	14.67, 14.67, 14.67
B ₂	16, 18, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21	21, 21, 21
B ₄	22, 25, 25	24	24, 24, 24
B ₅	25, 28, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	35, 35, 35	35	35, 35, 35
B ₈	38, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56	56, 56, 56

4) Use the two methods above to normalize the following
group of data: 200, 300, 400, 600, 1000 (a) min-max normalization
by setting min=0 and max=1 (b) z-score normalization

a) 200, 300, 400, 600, 1000

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$\text{Min} = 200$$

$$\text{Max} = 1000$$

Value

Normalizade

200

0.00

300

$$\frac{300 - 200}{800} = 0.125$$

400

$$\frac{400 - 200}{800} = 0.25$$

600

$$\frac{600 - 200}{800} = 0.50$$

~~1000~~

1000

$$\frac{1000 - 200}{800} = 1.00$$

b) $z = \frac{\lambda - \mu}{\sigma}$

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{500} = 500$$

value

$(x - \mu)$

$(x - \mu)^2$

200

-300

90,000

300

-200

40,000

400

-100

10,000

600

100

10,000

1000

500

250,000

$$\sum (x - \mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000} \approx 282.84$$

$$(200 - 500) / 282.84 = -1.06$$

$$(300 - 500) / 282.84 = -0.71$$

$$(400 - 500) / 282.84 = -0.35$$

$$(600 - 500) / 282.84 = 0.35$$

$$(1000 - 500) / 282.84 = 1.77$$

$$-1.06, -0.71, -0.35, 0.35, 1.77$$